DATA ENGINEERING PROJECT REPORT

Project Title: Discount Validator ETL Pipeline

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Abstract

The Discount Validator ETL Pipeline is a data engineering project designed to validate coupon applications in an e-commerce environment. The system ingests raw coupon data, processes it through multiple validation stages, and produces reliable analytical outputs. The pipeline follows a layered architecture similar to Bronze, Silver, and Gold layers. Raw data is preserved, cleaned, validated, and transformed into business-ready metrics. Key features include Dead Letter Queue (DLQ) handling for malformed records, conflict resolution between multiple coupons, and financial safeguards. The final output includes validated transactions, rejection reasons, and revenue insights.

Introduction

In modern e-commerce systems, discount coupons are widely used to attract customers and increase sales. However, improper validation of these coupons can lead to revenue leakage and inconsistent reporting. This project focuses on building a robust ETL pipeline to validate coupon applications efficiently.

Problem Statement

E-commerce platforms often face challenges such as duplicate coupon usage, expired coupons, incorrect category applications, and malformed data. Without a proper validation system, these issues lead to incorrect billing and financial losses. The project aims to automate coupon validation and ensure accuracy and consistency.

Objectives

• Validate coupon applications using business rules
• Detect and isolate malformed data using DLQ
• Resolve conflicts between multiple coupons
• Ensure accurate financial calculations
• Generate business insights and reports

Scope and Boundaries

The project focuses on batch processing of coupon data using Python and SQLite. It does not include real-time streaming, cloud deployment, or distributed systems. The pipeline is executed locally.

Technology Stack

Python – Pipeline development
Pandas – Data processing
NumPy – Numerical operations
SQLite – Data storage
Logging – Monitoring and debugging

System Architecture

Data flows through multiple stages:
1. Extraction – Load raw coupon data
2. Bronze Layer – Store raw data
3. Silver Layer – Clean and validate data
4. Gold Layer – Generate analytical outputs
5. Load – Store in SQLite
6. Reporting – Generate business insights

Bronze Layer

The Bronze layer stores raw coupon data with minimal transformation. It preserves original records and ensures data traceability.

Silver Layer

The Silver layer performs data cleaning and validation. It removes duplicates, checks for expired coupons, validates categories, and identifies malformed records.

Gold Layer

The Gold layer generates business-ready data. It calculates final payable amounts, applies valid discounts, and provides insights such as revenue impact and rejection statistics.

Data Quality Checks

- Null value check
- Negative price validation
- Duplicate detection
- Expiry validation
- Category mismatch detection

Results and Outputs

The pipeline successfully processes coupon data, identifies invalid records, and calculates correct payable amounts. It generates insights such as accepted vs rejected coupons and revenue impact.

Challenges and Solutions

Handling malformed data – Solved using DLQ
Conflict resolution – Solved using ranking logic
Performance optimization – Solved using vectorized operations

Conclusion

This project demonstrates a complete data engineering pipeline with validation, transformation, and reporting. It ensures data quality and provides meaningful business insights.

Future Scope

- Real-time processing using Kafka
- Dashboard integration
- Cloud deployment
- Automated scheduling using Airflow